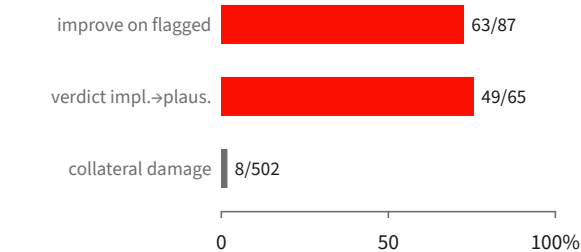


The loop helps on the critic's own metric, but an independent rater sees no change

PhyReAct's own physical metric

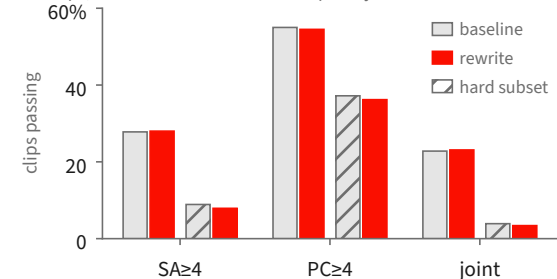
on the clips it flagged — the rewrite's objective



Left panel, whole set: mean $p_{plausible}$ 0.797 → 0.862 over all 589 clips (acted on 87/589, mean Δ +0.524 on those).

VideoPhy-2 AutoRater (independent)

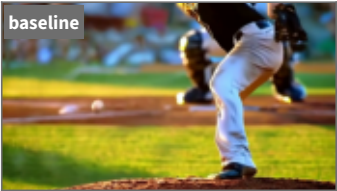
same clips, before vs after — not the loop's objective



Left: on its own metric the loop is targeted and low-collateral. Right: the independent rater registers no change — in either dimension, All or hard.

What the critic flagged and the rewrite regenerated

Ball appears from nothing → held in the hand



a ball floats in the field, no hand on it

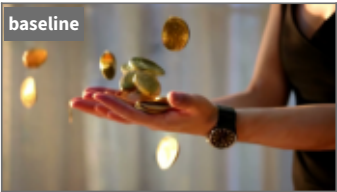
cited obligation: the baseball is present in the pitcher's hand at the start, not spawned in mid-air

rewriter added: "...the ball firmly gripped in the pitcher's fingers before the wind-up..."



the ball is in the pitcher's grip

Coins fuse into a blob → stay distinct discs



coins melt into one shapeless mass

cited obligation: coins keep their circular disc shape and do not morph or fuse

rewriter added: "...several distinct circular coins, each keeping its rigid disc shape..."



separate, clearly circular coins